

Meet in the Middle

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array of n numbers. In how many ways can you choose a subset of the numbers with sum x ?

Input

The first input line has two numbers n and x : the array size and the required sum.

The second line has n integers $t_1, t_2, \dots, t_n$: the numbers in the array.

Output

Print the number of ways you can create the sum x .

Constraints

- $1 \leq n \leq 40$
- $1 \leq x \leq 10^9$
- $1 \leq t_i \leq 10^9$

Example

Input	Output
4 5 1 2 3 2	3